

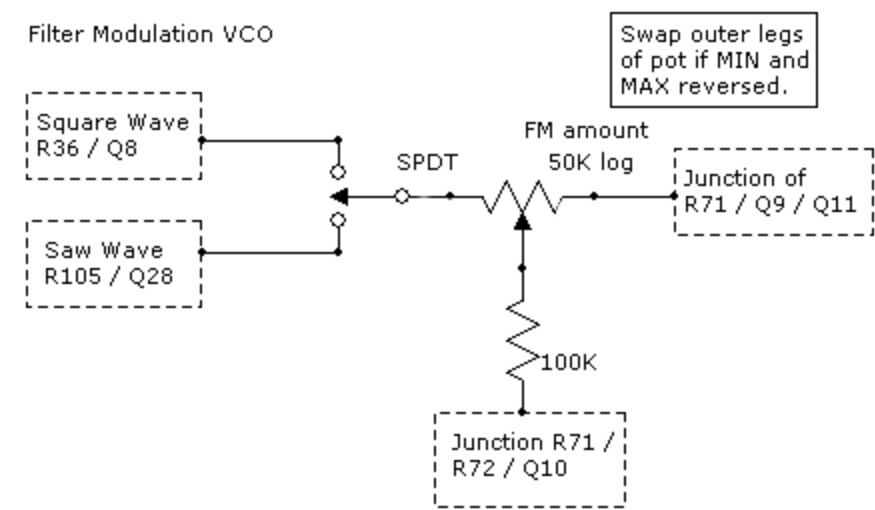
Filter Mods

Perhaps this page could be split out into a grouping of pages soon.

Filter Modulation (VCO)

This is another relatively easy but awesome mod for the 303. Take the feed from the square wave (R36/Q8) and the saw wave (R105/Q28) and wire one to pin3 of an SPDT switch and the other to pin1 of the same switch. Use a SPDT switch so that you can select the saw or square wave as the modulation source. Connect pin2 from the switch to the 3rd leg of a 50K log External Link [https://web.archive.org/web/20190104044840/http://example.com/]pot. Wire the first leg of the pot to the junction of R71/Q9/Q11 (it is best to hook it up here instead of to ground so that it follows the same bias strategy as the env mod does - see page 8 in the 303 service manual: http://dl.analoghell.com/index.cfm?id=107 [https://web.archive.org/web/20190104044840/http://dl.analoghell.com/index.cfm?id=107])). Hook up a 100K resistor to the center lug of the pot, the other end of the resistor goes to the junction of R71/R72/Q10-base, the summing point for the filter frequency control voltage input. This will let you select the saw or the square as the modulation source independent of the what you have selected for the VCO. Depending on where all the other controls are set you can get everything from an evil zipper to slowly dripping acid - among other things!

- **VERIFIED to work on the x0xb0x**

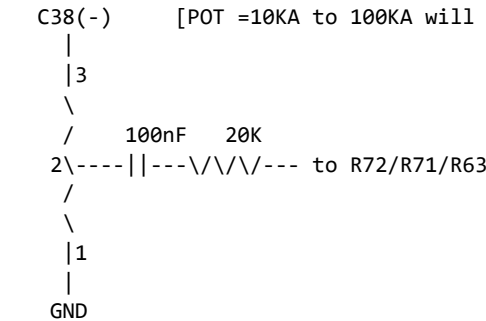


UPDATE: It is a good idea to buffer the waveform output. Use a 0.1uF ceramic cap after the 100K resistor. The Mod works great and is especially good at high resonance levels, so make sure you include the Resonance Boost Mod also!

UPDATE2 (by subatomic [https://web.archive.org/web/20190104044840/http://www.ladyada.net/forums/profile.php?mode=viewprofile&u=103]): that may be true (that 0.1uF being a good idea) however it doesn't sound as good...

Filter Modulation (VCA)

same as above. Instead this time you're taKing the signal from the VCA out (C38). I get the best result with the following...



Filter Overdrive

The DevilFish puts the filter into extreme duress. According to the engineer/artist you can overdrive the filter up to 66.6 times the original signal. R62 and C17 are responsible for attenuating the input into the VoltageControlledFilter from the VoltageControlledOscillator.

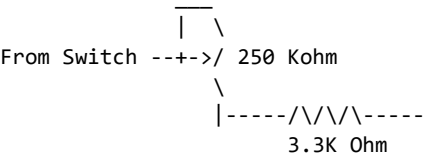
If you don't particularly care about the numerology or DevilFish purity, you can simply play with various resistor values until you find a sound that works. Conversely, you could mount a potentiometer instead.

C17 might attenuate the signal too much, so you might need to replace it with

a 10uF cap.

A Devilfish Like Overdrive

You could get a devilfish like overdrive if you remove R62,and put a 3.3K Ohm resistor in series with a 250K Ohm Pot. Note that this does not let you turn the input all the way off like the devilfish.



So what is this about Numerology?

I am not entirely sure about all of this, but R62 is a 220K resistor. If we want to overdrive the filter to 66.6 times, we need to decrease its value 66.6 times. 220 000 / 66.6 = 3,303.303 303 303

That is pretty groovy numerology, and probably explains the 'devil' in DevilFish.

Devilfish Overdrive Exactly

If you want to set up your overdrive like the devilfish so that it goes from 66x over keep some things in mind. The VCO output is from 5.33v to ~10.5v, so you should make sure you reference Pin 1 of the pot to the 5.33V bias voltage. You can take this from many places, such as IC16 Pin7/IC16 Pin3/C60(+). Pin 2 of the pot is wired between the waveform swith pin 2 and R62. Pin 3 goes to a 3.3K resistor and then to the other side of R62/C17(+). A 10KA, 50KA or 100KA pot will do.

I found an intesting fact recently while continuing the design of a custom 303 mod. The output from the VCO is not buffered. In the stock 303 this is not a problem because the signal is just attenuated and then ac-coupled to the input of the VCF. However, if you add a level control that references to the 5.333v bias voltage of the 303 - such as you would if you were adding a level control - there is a load effect on the VCO signal if it is not buffered first. The result is that the character of the square wave changes quite significantly. It looks sublt on a scope but it is very audible. So you're better off not referencing your overdrive pot to the 5.333v bias point. I know that the devilfish doesn't buffer the output of the VCO so it would be interesting to compare the square wave coming from a devilfish with the overdrive/level control in place versus with it bypassed.

Update Feb 1, 2007. I have added a sound clip so that people can hear what the 303 sounds like (specifically the square wave) when a filter overdrive control is used for which the first leg is reference to the 5.333v bias voltage. Specifically the case is when using a pot as a level control as in the Devilfish so that you can have the VCO input to filter range from off to severely overdriven. http://www.alphazone1.com/tb-303/303_filter_overdrive.mp3 [https://web.archive.org/web/20190104044840/http://www.alphazone1.com/tb-303/303_filter_overdrive.mp3] links to a sample of both a stock 303 and the same 303 with the overdrive control added as described above in the Wiki. The first 4 notes are the stock 303. The second 4 notes are the 303 with the overdrive control. Note the overdrive control is set for exactly the same output as the stock 303, so the level to the filter is exactly the same in both cases. The first comparision was done with Cutoff and Resonance at max. Next you will hear another 4 notes that is the stock 303 followed by another 4 notes that is again with the overdrive control in place. This time the cutoff control was set at max and the resonance control was set to mid range. On both these comparisions you can hear a fundamental difference in the sound of the square wave. Whether it's important or not is for you to decide. I just wanted to note that it's there. Aside from using a control like the overdrive pot discussed here *any* mod post-VCO and pre-VCF that references to the 5.333 bias voltage will result in some discoloration of the sound. The best thing to do is not to use a voltage divider but rather just current limit the signal into the VCF, simply use a 250k pot in series with a 3.3k resistor both in place of R62. This won't allow you turn turn the VCO signal all the way off (short of adding a switch) but it's not important for most people anyway

(note this text was recently deleted with no explanation)

Higher Beta Transistors on Q9 and Q10

This isn't strictly a mod per-sae, but is rather something you can do during fabrication to make sure you get the most 303-like sound. The original 303 used 2SA733ap, which has a beta gain of 300-350. The 733ap has been discontinued, but there is the 733p, which has a gain of about 280-310. The key is to test the beta of your transistors and find the highest beta transistors you can.

For more informastion, see HighGainTransistors

Resonance Warble

This is actually part of the [Power Supply Mods](#). However, it seems to be *not* working.

Increased Filter Low End

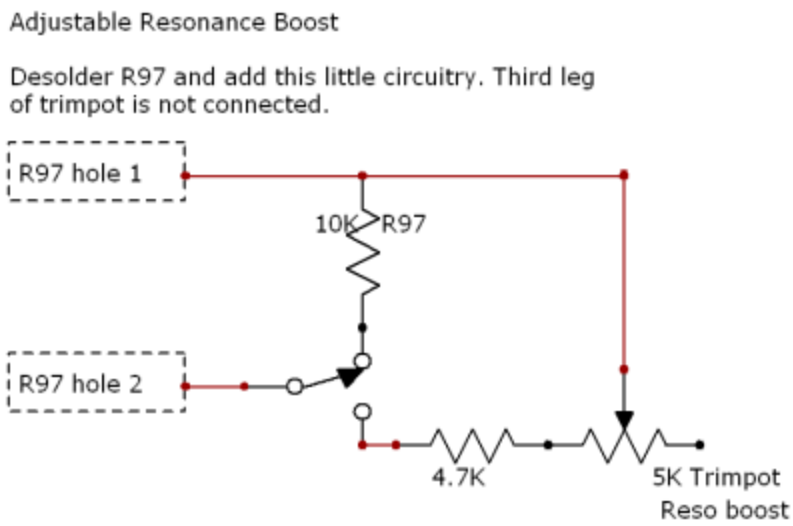
According to RobbinWhittle, all you have to do is short out R47 (10K ohm resistor) by a length of insulated wire. Some other x0xers tried it out, and found that mounting a 4.7K ParallelResistor along with R47 works better (giving 3197 ohms resistance). This will also affect cutoff range to a degree.

Resonance Boost

You can change the value of some of the resistors on the VoltageControlledFilter to boost the resonance. [The Mods page on Ladyada.net](#) [<https://web.archive.org/web/20190104044840/http://http://www.ladyada.net/make/x0xb0x/mods.html>] suggests:

1. Decrease R97 to ~7.5K
2. Decrease R108 to ~1.8K
3. Decrease R112 to ~5K
4. Increase R109 to ~22K (perhaps even larger?)

NOTE: You really don't have to bother with anything but decreasing R97. Changing it to 5K or 6K is about as low as you would want to go, and definitely a switch to choose between the factory setting of 10K and the new hi-res would be a good idea. An elegant way to do it is to use a 4.7K resistor in series with a 5K trimpot. This way you can adjust the maximum Reso boost to your liking without the need to solder and desolder different values of resistors.



Extra results: Momerath says in this post [<https://web.archive.org/web/20190104044840/http://forums.ladyada.net/viewtopic.php?t=5215>] "here are my suggestions for making a x0xb0x sound just like a 303" **r46 = jumper / r97 = 8.2k**. Presumably he added this as default (always on) on his box... Also he says "6.8k [for r97] was ridiculous if i recall". [charlie says confirms](#) it's good [<https://web.archive.org/web/20190104044840/http://forums.ladyada.net/viewtopic.php?t=5215&postdays=0&postorder=asc&start=15>] and says "you tend to get self resonance at the (very) top end but, I don't seem to have lost anything that I had before, and some combinations are definitely fuller. " and "but overall I think it's worth doing."

Subatomic [<https://web.archive.org/web/20190104044840/http://www.ladyada.net/forums/profile.php?mode=viewprofile&u=103>] tried mome's mod, and found that it can be a bit whistly, but good, I went with a pot though so I could tweak. To explain Mome's use of shorting out r46... It's the "[Resonance Increase Accent Range Mod](#)" [https://web.archive.org/web/20190104044840/http://www.ladyada.net/wiki/x0x/envelopemods#increase_accent_range] it seems to add more squelch on the accent, it's kinda subtle, but I could notice it most when the cutoff was way down it seemed to let through more resonant "scream" (sounds like "wow")...

VCF External In

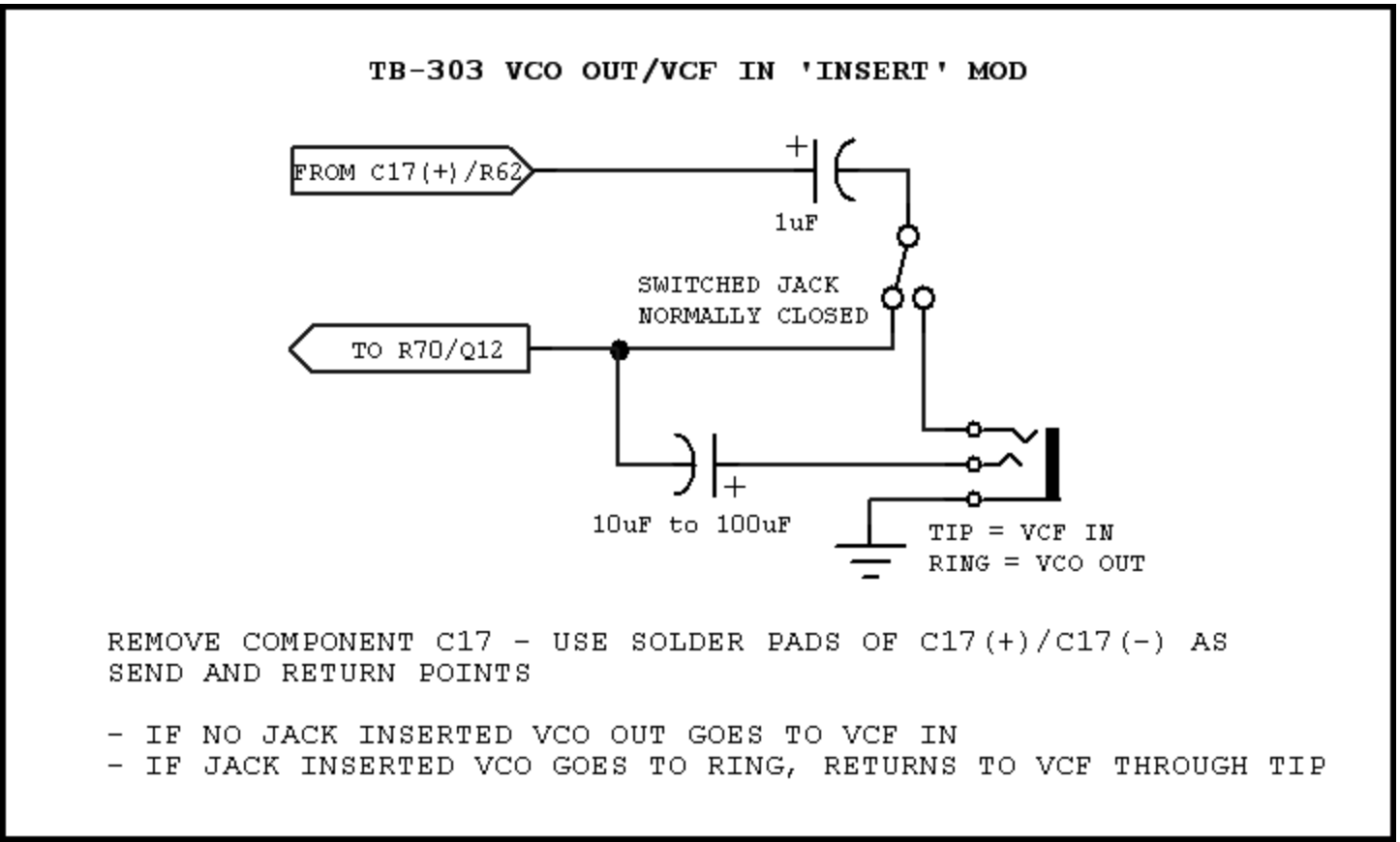
Updated Mod for VCO out/VCF in

This mod provides VCO output on Ring contact and VCF input on Tip contact of a single 1/4" TRS jack.

Cut trace between C17(-) and R70 [on upper side of mainboard]. Wire C17(-) to Ring contact of 1/4" TRS switching jack. Wire switching side of Ring contact back to R70/Q12 base junction. Connect Tip contact of jack to a 1uF 50V capacitor - use a bipolar electrolytic capacitor, or connect Tip contact to the positive lead if using a polarized cap. Connect the other/negative lead of the cap to the R70/Q12 base junction. **NOTE:** you may want to add a resistor in series between the Tip contact of the jack and the capacitor; otherwise the input may overdrive too easily. I experimented and found a 10K resistor gave a decent match between line-level input and the normal level of the oscillator. You could also put an input-

attenuating potentiometer in series here. This mod can be used with a y-cable to provide a pre-filter effects insert for your x0x, or you can plug in a regular instrument cable to mute the oscillator and run external audio through the filter, envelope generator and accents!

User [bcbox](#), known for his excellent x0xm0ds, wrote on the x0xb0xforums: *"use a 1uF cap and a 22k resistor - that gets you right around the input level you need and the 1uF is big enough to cover the frequency range of the x0x. Make sure you put the (+) side of the cap at the input to the VCF...so the tip of the jack goes to a 22k resistor then the resistor goes to the (-) side of the 1uF cap and the (+) side of the cap goes to R70/C17"*



From the Mods page on [ladayada.net](#)

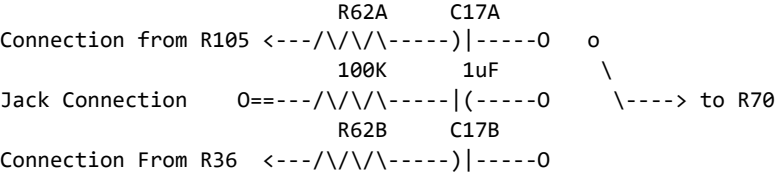
Use any panel mount jack, and connect the sleeve to ground. The tip should be connected to one side of a 100K resistor, the other side is connected to the negative side of a 1uF capacitor, and the positive side connects to the point where R70 and C17 touch.

Switched Input

VerifyMe: The switched input mod is untested, and is only theoretical. I may have screwed up on the polarity of the caps!

One could also use a 3 way switch on the waveform select inseat of a two way switch. In this case, you would still connect the tip to a 100K resistor, and then to the negative side of a 1uF capacitor.

You would need to remove R62 and C17, and perhaps put an equivalent 220K resistor and 1uF capacitor on the saw and square pole respectively. Note that the polarity between C17 and the 1uF capacitor that you add are different, so you cannot double up on them.



Note that R62A and R62B are seperate resistors, same with C17A and C17B being seperate capacitors.

You will also want to check on the Filter Overdrive mod, as doing this mod will affect it.

VCF Audio Out/VCA Audio In

VerifyMe: This mod is untested and unknown and purely theoretical. It also needs someone with more experience to have a peek at it

The signal where R121 and R122 meet is where the audio from the VCF comes out. One could in theory, wire a Jack between this and IC15. Hell, you could probably wire in a Tip/Ring/Sleeve Jack and provide both VCF out and VCA In functionality here. You may need to wire some value of resistor and/or capacitor before the jacks however!

'Fixing' the 'Flawed' Filter

VerifyMe: This mod is untested and unknown and purely theoretical. It also needs someone with more experience to have a peek at it

In theory by replacing the .018uF cap (C18) on the transistor ladder to a 0.033uF cap instead, you would make the filter a true 4 pole filter, and it would be drive itself to self-oscilation.

From a different webpage, apparently if you replace resistor R 97 (10k) with 6.8k (or 5.6k) it will allow the filter to self oscillate. → this is the Resonance Boost Mod!]]

CV for Cutoff

Connect a 100k resistor to the base of q10, and the other side of the resisitor would go to the tip of your CV jack. Note that there is no CvProtection.

There is also an issue with the voltage offset of the cutoff, because the signal is not referenced to ground, but is instead referenced to the emitter of q9 due to the way the envelope is applied to the signal. This is explained further in the VoltageControlledFilter section.

In practice this mod works very well, with the cutoff going between sub-sonically low to the normal maximum.

Filter Tracking

WorkOnMe: The details of this mod need to be fleshed out a little more

1:1 Filter tracking would be a simple matter. Just connect CV out into a 100k resisitor, and then into q10, like the CV for cutoff mod above. You will probably want to wire in a switch between there.

For DevilFish like filter tracking, this is a little more complex, as we need to scale the voltages. This would probably mean using a pot as part of a voltage divider, and possibly multiplying the CV in before going through the divider.

increase resonance feedback

reported by bcbox in this post [https://web.archive.org/web/20190104044840/http://forums.ladyada.net/viewtopic.php?t=5316], the following is a cut-n-paste of his post.

VerifyMe: Please remove this tag if you verify this mod works as described

you can increase resonance feedback by changing C21 and R122

Replace R122 with 68k Replace C21 with .015uF

this will increase resonance feedback while keeping the frequency response the same. The sound level will stay more constant as you increase resonance - not so much of a drop off.

Navigation

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Testing	Testing the Ps	Testing the Vco	Testing the Vcf	Testing the Envelope	Testing the Vca	Testing the Headphone And Mixer	Testing the Sequencer	Testing the Midi and Sync	
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